Brain Tumor Detection Model

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Abstract:

Brain tumor detection using medical imaging is a challenging task that often requires the expertise of radiologists. This project presents a deep learning–based solution capable of automatically identifying brain tumors from MRI scans. The model leverages convolutional neural networks (CNNs) trained on MRI images to achieve high accuracy in classification. The system is deployed using Streamlit, providing a simple, web-based interface that allows users to upload MRI images and receive predictions in real time. An additional image validation layer ensures that only MRI scans are processed, preventing invalid inputs from producing misleading results.

Introduction:

Brain tumors pose serious health risks, making early and accurate detection crucial for effective treatment. Traditional diagnostic processes are time-consuming and prone to subjectivity. Artificial intelligence (AI) and machine learning (ML) provide opportunities to improve diagnostic workflows through automated image analysis.

This project demonstrates how deep learning can be applied to brain MRI data for tumor detection. The final system integrates both machine learning experimentation and modern deployment practices, offering a practical, scalable solution suitable for research, education, and clinical assistance.

Methodology:

Technology Stack:

- **Deep Learning Frameworks**: TensorFlow, Keras
- **Data Handling**: NumPy, Pandas
- **Image Processing**: PIL (Python Imaging Library)
- **Visualization**: Altair

- **Deployment**: Streamlit

Model Development:

The development process involved several stages:

1. **Baseline Models**
   Logistic Regression and Random Forest were tested initially to establish a benchmark. XGBoost was also evaluated, showing improvement on structured data but limited effectiveness for image-based classification.

2. **Deep Learning Approach**
   A convolutional neural network (CNN) was implemented using TensorFlow and Keras. The CNN outperformed traditional models, achieving strong results on grayscale MRI images. The model architecture was trained to classify between tumor-present and tumor-absent cases, leveraging image preprocessing techniques for consistency.

3. **MRI Validation Layer**
   Since end-users might upload non-MRI images, a validation function was designed to reject invalid inputs. This step relies on grayscale variance, mean intensity, and other heuristics to filter out irrelevant images. Only valid MRIs proceed to prediction.

Workflow:

1. **Upload**: User uploads an image.

2. **Validation**: Image undergoes MRI validation.

3. **Preprocessing**: Images are resized, normalized, and formatted for CNN input.

4. **Prediction**: The CNN generates tumor presence/absence classification.

5. **Output**: Results are displayed in real time through Streamlit, alongside the uploaded image.

Deployment:

Deployment was achieved via **Streamlit Cloud**. The trained model (`.h5` file) and application code were hosted in a GitHub repository, enabling smooth integration with Streamlit. Dependencies were managed via a `requirements.txt` file, ensuring reproducibility across environments.

The Streamlit interface was optimized for both desktop and mobile views, with responsive design elements to ensure accessibility.

Applications:

- **Medical Assistance**: Supports radiologists in diagnosis by providing rapid preliminary results.
- **Education**: Serves as a project template for students learning deep learning in healthcare.
- **Research**: Provides a foundation for further experimentation with medical imaging datasets.

Conclusion:

The Brain Tumor Detection Model highlights the power of deep learning in medical diagnostics. Through iterative development—starting with classical ML algorithms and culminating in CNN-based architecture—the project demonstrates significant accuracy improvements. Its deployment on Streamlit Cloud ensures real-time accessibility, while the MRI validation layer enhances the reliability of predictions.

This system showcases how AI can bridge the gap between research and practical application, offering potential as a clinical decision-support tool with further refinements.

References

- Kaggle Dataset: Brain Tumor MRI Dataset
- TensorFlow Documentation: https://www.tensorflow.org
- Streamlit Documentation: https://docs.streamlit.io